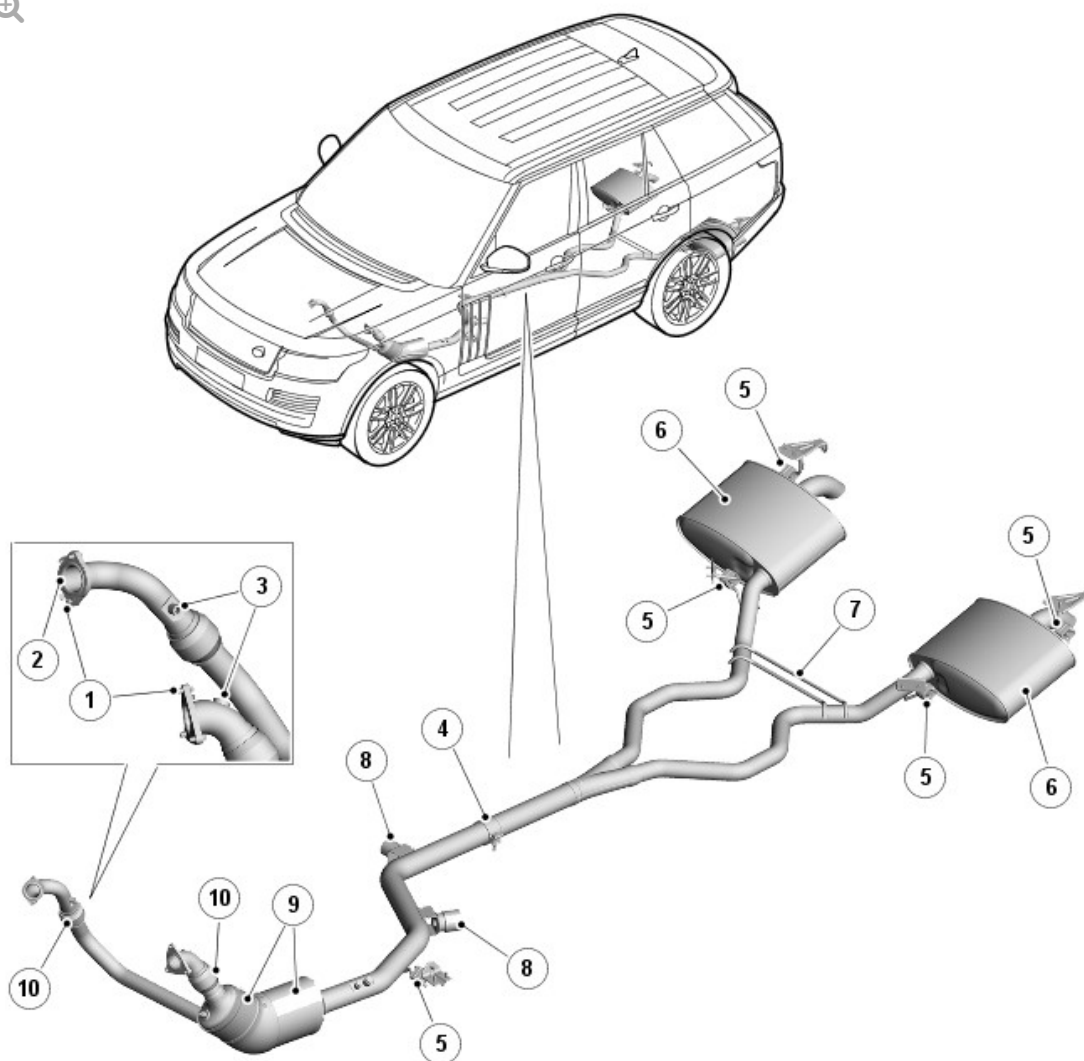


2016.0 RANGE ROVER (LG), 309-00

EXHAUST SYSTEM – TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

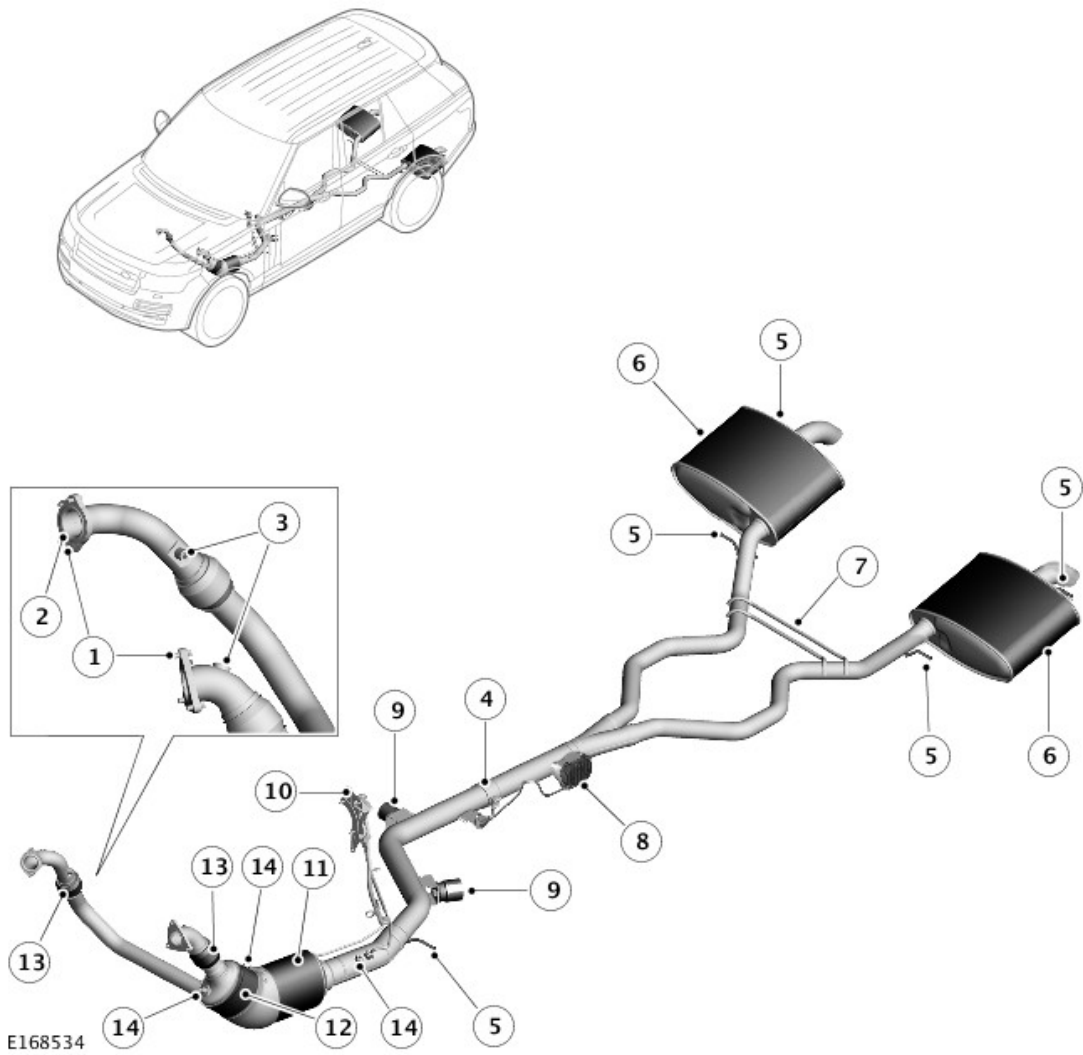
TDV6 3.0L DIESEL – BI-TURBOCHARGER
EXHAUST SYSTEM COMPONENT
LOCATION – WITHOUT DIESEL
PARTICULATE FILTER



E147624

ITEM	DESCRIPTION
1	Bolts (5 Off)
2	Gasket
3	HO2S Mounting Boss
4	Clamp
5	Mounting Rubber (5 Off)
6	Silencer - Rear
7	Exhaust Brace
8	Mass Damper (2 Off)
9	Catalytic Converter
10	Flexible Coupling

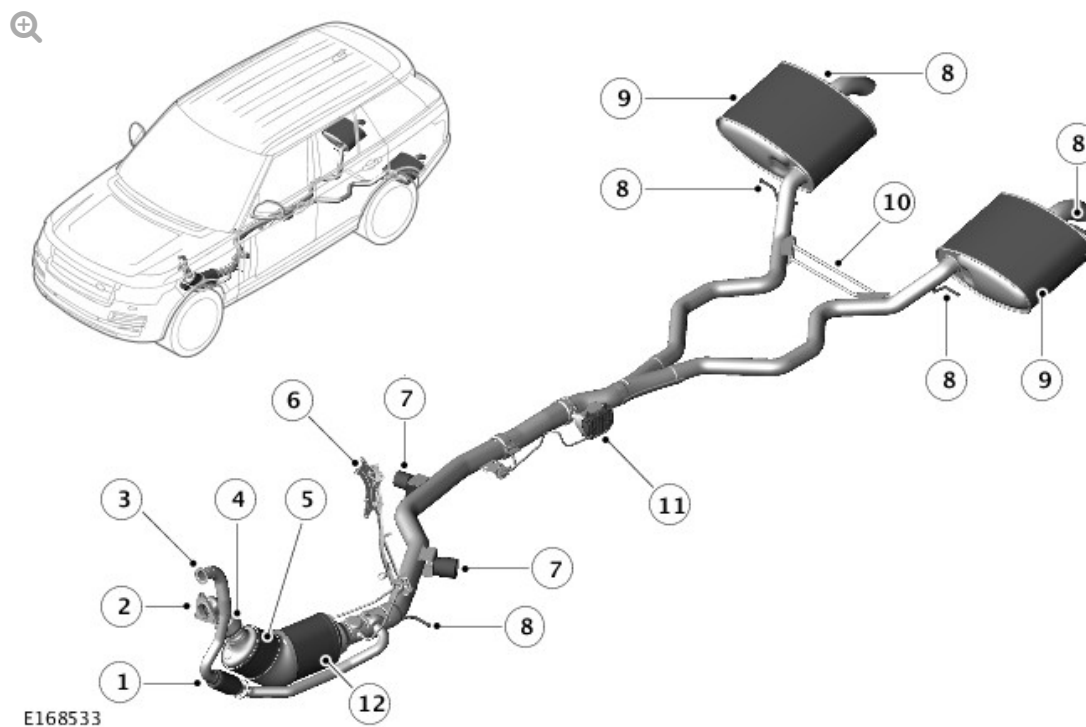
TDV6 3.0L DIESEL - BI-TURBOCHARGER
EXHAUST SYSTEM COMPONENT
LOCATION - WITH DIESEL PARTICULATE
FILTER



ITEM	DESCRIPTION
1	Bolts (5 Off)
2	Gasket
3	HO2S Mounting Boss
4	Clamp
5	Hanger bar

6	Silencer - Rear
7	Exhaust Brace
8	Exhaust Valve
9	Mass Damper (2 Off)
10	Differential Pressure Sensor
11	Diesel Particulate Filter
12	Catalytic Converter
13	Flexible Coupling
14	Exhaust Gas Temperature Sensor Mounting Boss

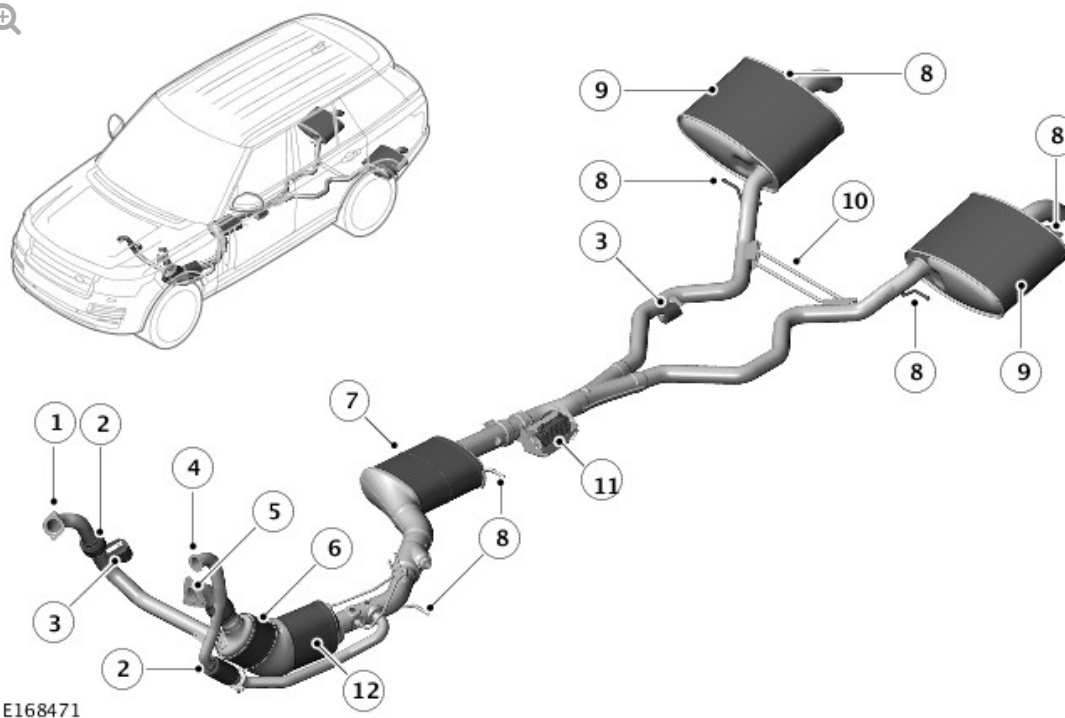
TDV6 3.0L DIESEL - SINGLE TURBOCHARGER EXHAUST SYSTEM COMPONENT LOCATION - WITH DIESEL PARTICULATE FILTER



ITEM	DESCRIPTION
1	Flexible Coupling
2	Flange connection - Turbocharger

3	Flange connection – Low Pressure Exhaust Gas Recirculation (LPEGR) valve
4	Flexible Coupling
5	Catalytic Converter
6	Differential Pressure Sensor
7	Mass damper
8	Hanger bar
9	Rear silencer
10	Exhaust brace
11	Exhaust valve
12	Diesel Particulate Filter (DPF)

COMPONENT LOCATION - BI-TURBOCHARGER VEHICLES WITH SELECTIVE CATALYST REDUCTION SYSTEM

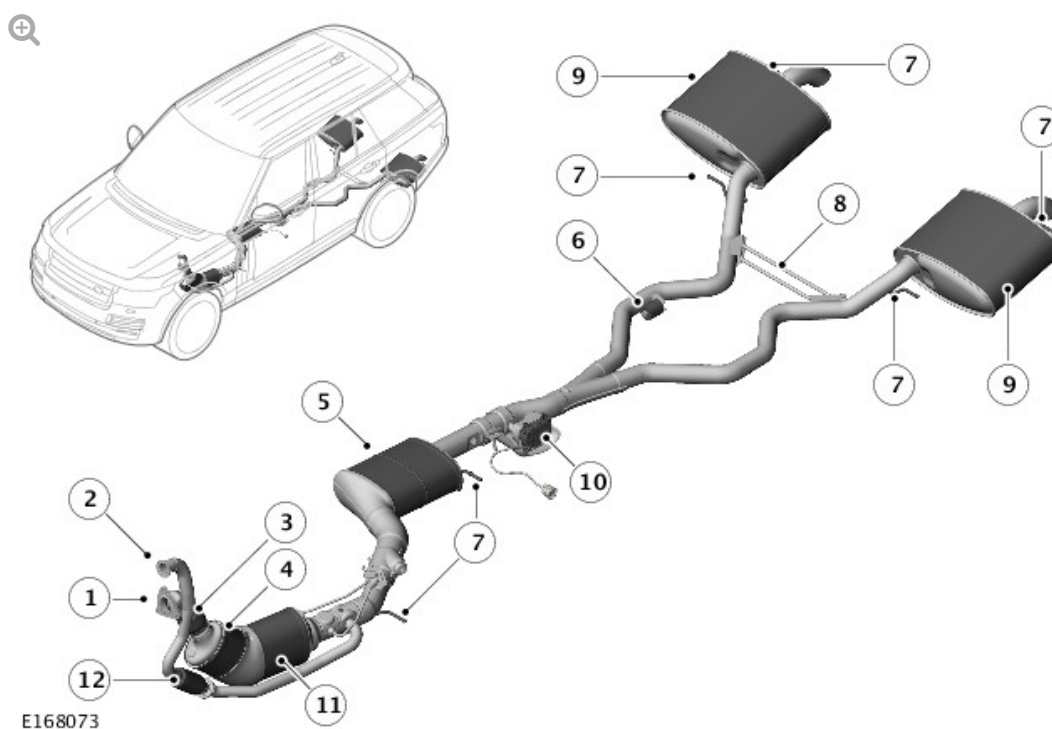


ITEM

DESCRIPTION

1	Flange connection - Secondary turbocharger
2	Flexible coupling
3	Mass damper
4	Flange connection – Low Pressure Exhaust Gas Recirculation (LPEGR) valve
5	Flange connection - Primary turbocharger
6	Catalytic converter
7	Selective Catalytic Reduction (SCR) catalytic converter
8	Hanger bar
9	Rear silencer
10	Exhaust brace
11	Exhaust valve
12	Diesel Particulate Filter (DPF)

COMPONENT LOCATION - SINGLE TURBOCHARGER VEHICLES WITH SELECTIVE CATALYST REDUCTION SYSTEM



ITEM	DESCRIPTION
1	Flange connection - Turbocharger
2	Flange connection – Low Pressure Exhaust Gas Recirculation (LPEGR) valve
3	Flexible coupling
4	Catalytic converter
5	Selective Catalytic Reduction (SCR) catalytic converter
6	Mass damper
7	Hanger bar
8	Exhaust Brace
9	Rear silencer
10	Exhaust valve
11	Diesel Particulate Filter (DPF)
12	Flexible coupling

OVERVIEW

There are five variants that are available on the TDV6 3.0L Exhaust System:

- Non DPF System – EU4 Emission regulations.
- Bi-Turbocharger DPF System – EU5 Emissions regulations.
- Single turbocharger DPF System – EU5 Emissions regulations.
- Bi-Turbocharger DPF System with SCR System – EU6 Emission regulations.
- Single turbocharger DPF System with SCR System – EU6 Emission regulations.

On vehicles with a DPF fitted, a single catalytic converter is located in the bank 2 exhaust downpipe from the manifold. On vehicles without a DPF, a second catalytic converter is fitted in place of the DPF.

On vehicles with SCR System fitted, an additional SCR catalytic converter is located in the exhaust downpipe after the DPF.

The system is attached to the underside of the body with 5 mounting rubbers which are located on hanger brackets that are welded to the system.

The mounting rubbers locate on corresponding hanger brackets which are bolted to the underside of the vehicle body.

CAUTIONS:

- The use of bio fuels can seriously contaminate and destroy the coatings used on the catalytic converter. The DPF and the catalytic converter can become irreversibly contaminated if non-specified oils or fuels are used. This can result in the vehicle being unable to regenerate the DPF, becoming non-compliant with the rear silencers emission regulations and replacement of the catalytic converter and DPF will be required.
- If the vehicle is waded in deep water and the engine is stopped with the rear silencers submerged, the water, which can enter the system, can also contaminate both the DPF and catalytic converter. This can result in catalytic converter damage and damaging the ability for the DPF to regenerate therefore requiring both components to be replaced.

DESCRIPTION

FRONT SECTION - BI-TURBOCHARGER VEHICLES

The front section comprises of two catalytic converters or, if a DPF is fitted, one catalytic converter and diesel particulate filter. The bank 2 inlet pipe connects to the catalytic converter and bank 1 inlet pipe will connect to a half shell that is located between the two catalytic converters or catalytic

converter and diesel particulate filter (if DPF fitted).

On vehicles with SCR System fitted, an additional SCR catalytic converter is located in the front section after the DPF.

Each inlet pipe is fitted with a flange that connects to the turbocharger for each cylinder bank. The bank 2 inlet flange uses a 3 bolt fixing and the bank 1 inlet flange uses a 2 bolt fixing that are screwed into threaded holes in each of the turbochargers. Each flange is sealed using a gasket and both inlet pipes are fitted with flexible couplings.

The catalytic converters or DPF (if fitted) have an outlet pipe that is secured by a clamp to the centre section. The outlet pipe has 2 mass dampers which will absorb resonance from the system. The outlet pipe also has one hanger bracket which is welded to the pipe.

FRONT SECTION - SINGLE TURBOCHARGER VEHICLES

The front section comprises of one catalytic converter and a diesel particulate filter.

On vehicles with SCR System fitted, an additional SCR catalytic converter is located in the front section after the DPF.

The inlet pipe is fitted with a flange that connects to the turbocharger. The inlet flange uses a 3 bolt fixing that is screwed into threaded holes in each of the turbochargers. The flange is sealed using a gasket and the inlet pipe is fitted with a flexible coupling.

The DPF has an outlet pipe that is secured by a clamp to the centre section. The outlet pipe also has one hanger bracket which is welded to the pipe.

REAR SECTION

The rear section comprises a single pipe, which separates into two pipes that connect to the rear silencers. The rear section is connected to the front section with a clamp.

The two rear silencers are attached to the body by two exhaust hanger

brackets on each of the rear silencers.

A brace is welded between the pipes to retain system in position and reduce exhaust flexing.

CATALYTIC CONVERTER

The oxidizing catalytic converter is located in the front section of the bank 2 inlet pipe, after the pre-catalyst exhaust temperature sensor (vehicles with DPF only) and the Heated Oxygen Sensor (HO2S).

NOTE:

On vehicles fitted with DPF: The Pre and Post DPF Temperature Sensors are used by the engine management system to monitor the DPF regeneration purposes.

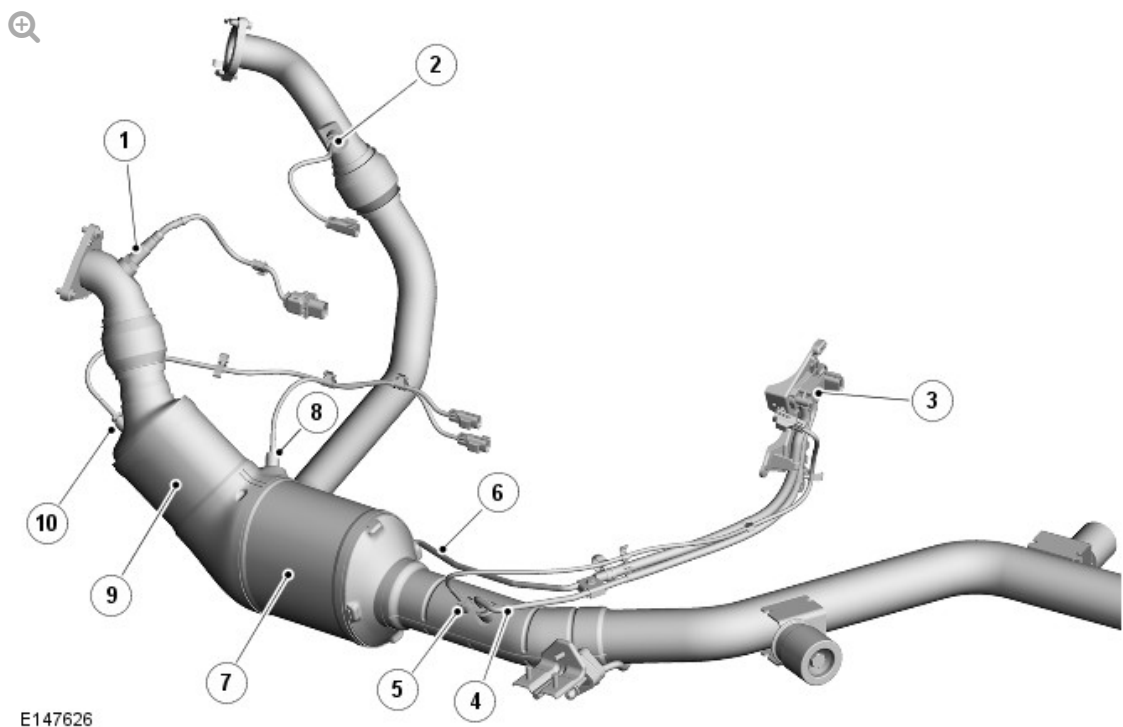
On vehicles without a DPF, a second catalytic converter is fitted in place of the DPF.

The engine management system provides accurately metered quantities of fuel to the combustion chambers to ensure the most efficient use of fuel and to minimise the exhaust emissions. On vehicles without a DPF, the second catalytic converter is fitted to further reduce the carbon monoxide and the hydrocarbon content of the gases. In the catalytic converter the exhaust gases are passed through honeycombed ceramic elements coated with a special surface treatment called 'washcoat'. The 'washcoat' increases the surface area of the ceramic elements by a factor of approximately 7000. The 'washcoat' is a coating containing platinum and palladium, which are active constituents for converting harmful emissions into inert by-products. The platinum and palladium add oxygen to the carbon monoxide and the hydrocarbons in the exhaust gases, and converts them into carbon dioxide and water respectively.

DIESEL PARTICULATE FILTER

NOTE:

Bi-Turbocharger system shown, single turbocharger system is similar.



E147626

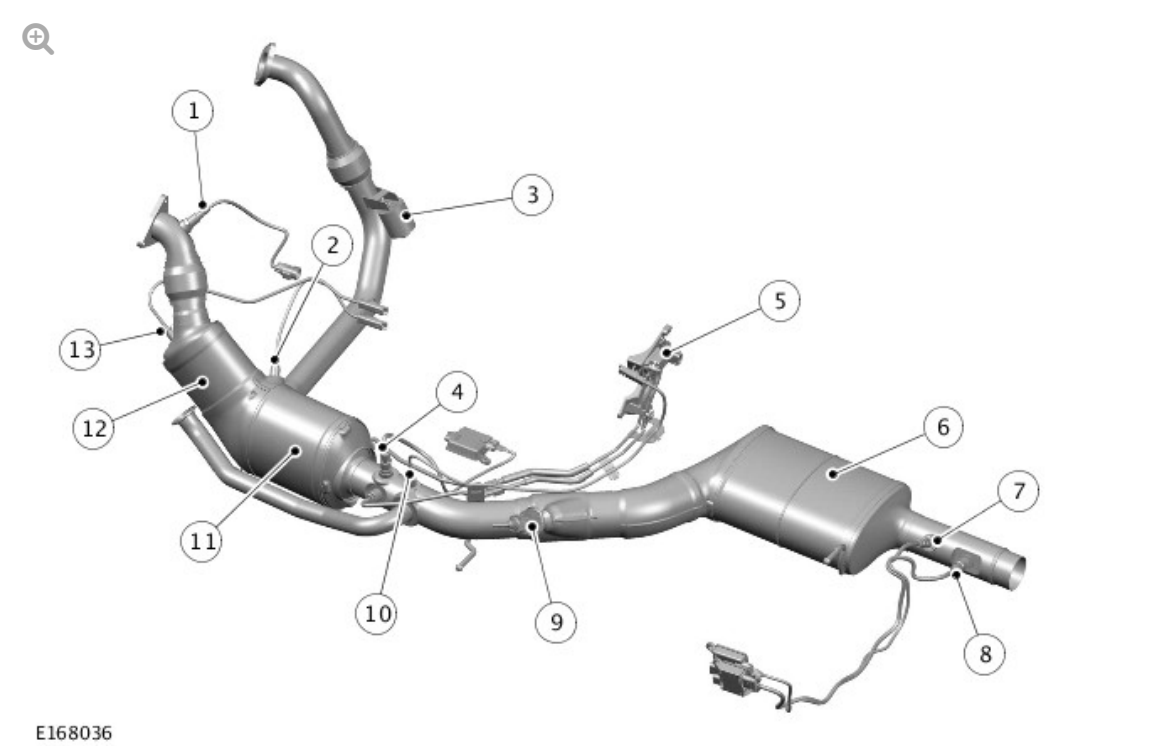
ITEM	DESCRIPTION
1	Heated Oxygen Sensor
2	Pre-DPF Temperature Sensor
3	Differential Pressure Sensor
4	Low Pressure Pipe
5	Post-DPF Temperature Sensor
6	High Pressure Pipe
7	Diesel Particulate Filter
8	Post-Catalyst Temperature Sensor
9	Catalytic Converter
10	Pre-Catalyst Temperature Sensor

For additional information, refer to: Diesel Particulate Filter (309-00 Exhaust System - TDV6 3.0L Diesel, Description and Operation).

SELECTIVE CATALYST REDUCTION SYSTEM

NOTE:

Bi-Turbocharger system shown, single turbocharger system is similar.



E168036

ITEM	DESCRIPTION
1	Heated Oxygen Sensor (HO2S)
2	Pre DPF exhaust gas temperature sensor
3	Mass damper (Bi-Turbocharger vehicles only)
4	Pre Selective Catalyst Reduction NOx sensor
5	Differential pressure sensor
6	Selective Catalyst Reduction (SCR) catalytic converter
7	Post Selective Catalyst Reduction soot sensor
8	Post Selective Catalyst Reduction NOx sensor

9	DEF Dosing Module
10	Post DPF exhaust gas temperature sensor
11	Diesel Particulate Filter (DPF)
12	Catalytic converter
13	Pre-Catalyst Temperature Sensor

For additional information, refer to: Selective Catalyst Reduction (SCR) (309-00A, Description and Operation).